VU Quiz Collection

Total Questions: 10 from 10 Quiz(es)

Courses: MTH301

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Waseem Akram | wasii.dev | +1 (226) 798-2360

Quiz 1 - MTH301 (Q1/10)

Question:

Which integral gives the arc length of the curve
r(t)=13t3i+tj+t2kr(t)=13t3i+tj+t2kr(t) = \frac{1}{3}{t^3}i + tj + {t^2}k over the interval [1,3]

Options:

A. "+13(1+t2)2"""""""" dt"+13(1+t2)2dt\int\limits_1^3 {\sqrt {{{(1 + {t^2})}^2}}

B. "+13(13+t)2"""""""" dt"+13(13+t)2dt\int\limits_1^3 {\sqrt {{{(\frac{1}{3} + t)

C. "+13(1+t)2"""""""" dt"+13(1+t)2dt\int\limits_1^3 {\sqrt {{{(1 + t)}^2}} } dt

D. "+131+t""""" dt"+131+tdt\int\limits_1^3 {\sqrt {1 + t} } dt

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Quiz 2 - MTH301 (Q2/10)

Question:

Graph of c=x0i+y0j+z0kGraph of c=x0i+y0j+z0kGraph~ of~ c = {x_0}i + {y_0}j + {z_0}k

Options:

A. None of these

B. is the curve passing through the point (x0,y0,z0)(x0,y0,z0)({x_0}, {y_0}, {z_0})

C. is the line passing through the point (x0,y0,z0)(x0,y0,z0)({x_0}, {y_0}, {z_0})

D. is the point (x0,y0,z0)(x0,y0,z0)({x_0}, {y_0}, {z_0})

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Quiz 3 - MTH301 (Q3/10)

Question:

If the integration is carried out along the path of a particular curve, such an integral is called . . . . . .

Options:

A. Line integral.

B. Indefinite integral.

C. Surface integral.

D. Volume integral.

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Quiz 4 - MTH301 (Q4/10)

Question:

The arc length of the portation of the circular helix where and and , then the arc lenght is

Options:

A. L = \int\limits_0^\pi {\sqrt 2 } dxL = \int\limits_0^\pi {\sqrt 2 } dx

B. L = \int\limits_0^\pi { - \sqrt 2 } dtL = \int\limits_0^\pi { - \sqrt 2 } dt

C. L = \int\limits_0^\pi {\sqrt 2 } dyL = \int\limits_0^\pi {\sqrt 2 } dy

D. L = \int\limits_0^\pi {\sqrt 2 } dtL = \int\limits_0^\pi {\sqrt 2 } dt

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Quiz 5 - MTH301 (Q5/10)

Question:

The vector is continuous at if

Options:

A. (a) is defined

B. (d) All (a), (b) and (c) .

C. (b) exist

D. (c)

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Quiz 6 - MTH301 (Q6/10)

Question:

If a scalar field exists for all points on the curve, the with , defines the line integral of i .e, line integral =

Options:

A. " Á=0" V(r)drÁ" Á=0" V(r)drÁ\sum\limits_{\rho = 0}^\infty {V(r)d{r_\rho }}

B. " Á=0nV(r)drÁ" Á=0nV(r)drÁ\sum\limits_{\rho = 0}^n {V(r)d{r_\rho }}

C. " Á=1nV(r)drÁ" Á=1nV(r)drÁ\sum\limits_{\rho = 1}^n {V(r)d{r_\rho }}

D. " Á=1" V(r)drÁ" Á=1" V(r)drÁ\sum\limits_{\rho = 1}^\infty {V(r)d{r_\rho }}

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Quiz 7 - MTH301 (Q7/10)

Question:

If r is a vector-valued function in 2-space or 3-
space, then
ddt["+r(t)dt]ddt["+r(t)dt]{\frac{d}{{dt}}\left[{\int {r(t)dt} } \right]}

Options:

A. "+r(t)dt"+r(t)dt{\int {r(t)dt} }

B. d"+r(t)d"+r(t)d\int {r(t)}

C. r(t)r(t){r(t)}

D. None of above

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Quiz 8 - MTH301 (Q8/10)

Question:

A line integral naturally involved --------- independent variables in space.

Options:

A. five

B. four

C. two

D. three

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Quiz 9 - MTH301 (Q9/10)

Question:

The differential equation is an exact differential equation .

Options:

A. True

B. False

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Quiz 10 - MTH301 (Q10/10)

Question:

Consider the two functions, and have continuous partial derivatives in a certain domain (say) . The differential equation, is an exact differential equation if and only if

Options:

A. " Q" x" Q" x\frac{{\partial Q}}{{\partial x}}

B. " P" y"`" Q" x" P" y"`" Q" x\frac{{\partial P}}{{\partial y}} \ne \frac{{\partial Q}}{{

C. " P" y =" Q" x" P" y =" Q" x\frac{{\partial P}}{{\partial y}} = \frac{{\partial Q}}{{\

D. " P" y" P" y\frac{{\partial P}}{{\partial y}}

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